Solving Systems by Elimination with Scaling

Algebra 1 Linear Systems and Modeling

Scaling one equation, scaling both equations, and building a system from a real situation before eliminating

The organizer. Elimination needs one variable's coefficients to be opposites (or equal). When they are not, *scale*: multiply every term of an equation — both sides — by a constant. Use this layout on every problem.

Step	**What you write**	**Why**
Target	"eliminate y"	pick the variable that is cheaper to match
Scale	multiplier next to each equation	make the coefficients opposites
Combine	add the scaled equations	the target variable drops out
Back-substitute	put the value into either original	find the second variable
Check	test the pair in *both* originals	a pair that fits only one is not a solution

The order of work for every problem on this sheet. No values shown: use the numbers given in each problem.

Directions.

- Write the multiplier you use beside each equation you scale.
- Solve for both variables and write the ordered pair on the answer line.
- Check your pair in both original equations before you write it down.

1. A school store sells notebooks and binders. Two notebooks and one binder cost \$11. One notebook and three binders cost \$18. Let x be the price of a notebook and y the price of a binder, both in dollars.

$$2x + y = 11 \qquad\qquad x + 3y = 18$$

Scale one equation, eliminate a variable, and find both prices.

Answer: ________________

2. Solve by elimination. Scaling one equation is enough here.

$$4x - 3y = 5 \qquad\qquad x + y = 3$$

Answer: ________________

3. A theater sold 12 tickets to one show for \$88 in all. Adult tickets cost \$9 each and child tickets cost \$5 each. Let a be the number of adult tickets and c the number of child tickets.

$$a + c = 12 \qquad\qquad 9a + 5c = 88$$

Solve the system by elimination and state how many of each ticket were sold.

Answer: ________________

4. Solve by elimination. Neither pair of coefficients matches, so both equations must be scaled.

$$3x + 4y = 10 \qquad\qquad 5x + 6y = 16$$

Answer: ________________

5. Solve by elimination.

$$5x + 2y = 4 \qquad\qquad 3x + y = 3$$

Answer: ___________________

6. A bakery sold 50 items on Saturday and took in \$172. Muffins sell for \$3 each and scones for \$4 each. Let m be the number of muffins and s the number of scones.

$$m + s = 50 \qquad\qquad 3m + 4s = 172$$

Solve the system by elimination and state how many of each were sold.

Answer: ___________________

7. Solve by elimination. Choose the variable that needs the smaller multipliers, and say which one you chose.

$$4x + 5y = 7 \qquad\qquad 6x + 7y = 11$$

Answer: ________________

8. A canoe travels 24 kilometres downstream in 2 hours and the same 24 kilometres back upstream in 3 hours. Let b be the canoe's speed in still water and c the speed of the current, both in kilometres per hour. Downstream the speeds add, upstream they subtract:

$$2b + 2c = 24 \qquad\qquad 3b - 3c = 24$$

Solve the system by elimination and state both speeds in kilometres per hour.

Answer: ________________

9. Find and fix the mistake. Maya is solving

$$x + 4y = 14 \qquad 3x - 2y = 14$$

She writes: "Multiply the first equation by 3 to match the $3x$: that gives $3x + 12y = 14$."
(a) Explain in one or two sentences what is wrong with her scaled equation and why multiplying *every* term by 3 leaves the solution unchanged.
(b) Scale correctly and solve the system.

Answer: __________________

10. A lab technician mixes a 30% salt solution with a 50% salt solution to make 20 litres of a 45% salt solution. Let x be the litres of the 30% solution and y the litres of the 50% solution. The volumes add, and the salt adds:

$$x + y = 20 \qquad\qquad 30x + 50y = 45 \cdot 20$$

Solve the system by elimination and state how many litres of each solution are used.

Answer: ________________